

Open science in stroke research

A living benchmark, 2015–2025

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Conflict-of-interest disclosure

Disclosures

- [x] No, nothing to disclose
- [ ] Yes, please specify

Company name	Honoraria / expenses	Consulting / advisory board	Funded research	Royalties / patent	Stock options	Ownership / equity position	Employee	Other (please specify)
None								

01 Background

Hybrid discipline



Neuroscience

Lesion–symptom mapping, disconnection, connectomes

Neurology

Trials, outcomes, recovery

Cardiology

Atrial fibrillation, PFO, vascular risk

The benchmark compares stroke with the same three fields

02 Open science

Purpose

1 Reproducibility

2 Access

3 Value

Reproducibility

Stroke and its neighbours

Neuroscience

200+

lesion network mapping studies, challenged in 2026 with public data alone

Analysis choices

Neurology

8 years

before the NINDS rt-PA data could be checked independently; the benefit held

Closed data

Cardiology

43%

higher odds of MI on rosiglitazone, in results the maker had to post

Unpublished results

Preclinical stroke

10 of 525

animal studies reported no effect

Publication bias

van den Heuvel 2026 · Ingall 2004 · Nissen 2007 · Sena 2010

04 Reproducibility · Access · Value

NINDS rt-PA

1995

Published

624 patients; benefit at 3 months

2003

Released

8 years later; US$79 CD, no codebook

2004

Re-analysed

Benefit holds (OR 2.1); BP data unusable

2009–23

Debated

Critics and trialists, same file

2026

Closed

“Contact the Principal Investigators”

NINDS 1995 · Dachs 2008 · Ingall 2004 · UK MHRA 2015 · NINDS archive, 15 Oct 2026

Access

Readers

+89% downloads when randomised to open access

1.6× citations, open vs closed stroke articles

Tools

2.8% thrombectomy access, median country

£15–30k a year per centre, imaging AI licence

Free DeepISLES, validated on 1,685 scans

> No open stroke tool is yet regulated or tested in low- and middle-income countries.

Davis, BMJ 2008 · this benchmark, 2015–2023 · MT-GLASS, Circulation 2023 · NICE 2024 · de la Rosa, Nat Commun 2025

Value

Waste	Reuse	Saved	Lost
85%	209	$0.9–1.7B	20–30%
of research investment, estimated	outside papers from 336 shared trials	to collect again the data behind 913 papers	less follow-on research, licensed genes

Chalmers and Glasziou, Lancet 2009 · Taherifard, BMJ 2026 · Milham, Nat Commun 2018 · Williams, J Polit Econ 2013

07 Benchmark

Method

> Audits exist for neurology, cardiology and medicine. **None for stroke.**

Retrieve	OpenAlex, stroke and its three neighbours, 2015 to 2025
Clean	Articles and reviews with a DOI
Detect	Open access, links, preprints, registration
Read	Up to 1,500 full texts a year
Publish	Code, data, dashboard; every six months

What it measures

- Open access
- Data and code
- Preprints
- Registration

140,281
stroke articles and reviews

16,362
stroke full texts read

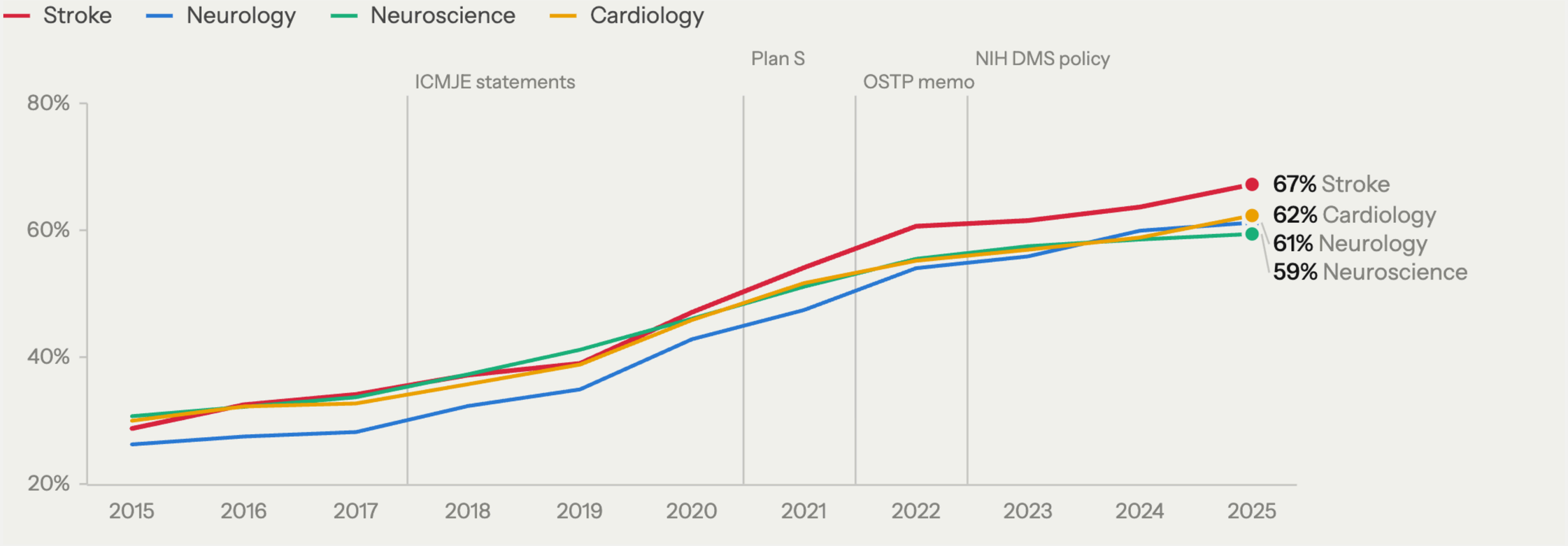
4
fields

08 Finding 1

Open access

29% → 67% published open, stroke, 2015 to 2025

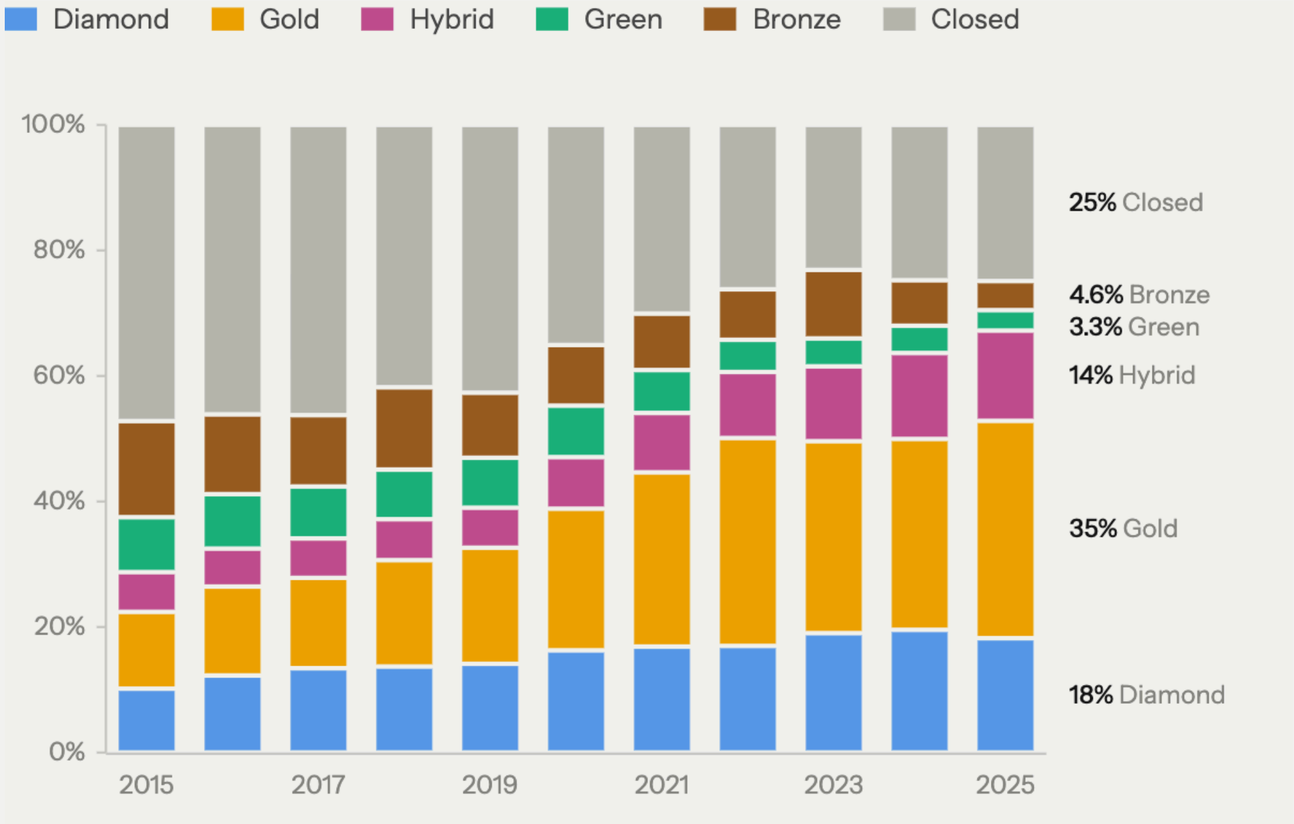
75% free to read by any route, 2025



Published open = gold, diamond or hybrid in OpenAlex · free to read = any status other than closed

09 Finding 2

Open-access routes



+18 points

gold, a paid route, 2015 to 2024

9% → 3%

green, a repository copy, 2015 to 2025

Share of stroke articles by OpenAlex status · 2025: counts complete; open-access status still settling

10 Finding 3

Data and code

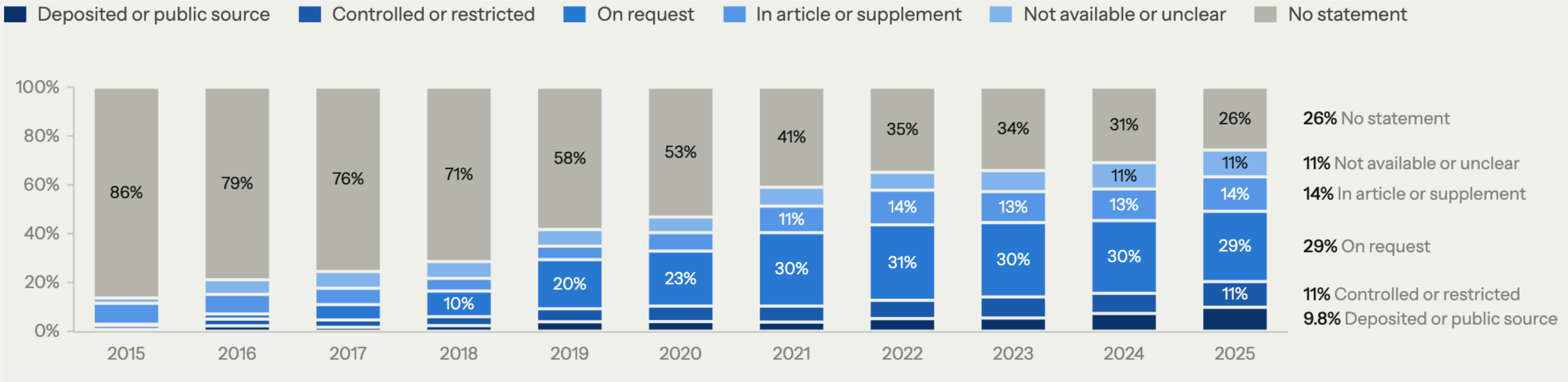
Data in a repository	Code link	Preprint version	Registration ID
0.2%	<0.1%	2.7%	1.5%
in 2024; never above 0.6%	every year since 2015	in 2024	of non-trial studies in 2024; trials 35%

Abstracts and metadata, 2015 to 2024: lower bounds

11 Finding 4

Data-availability statements

Write a statement	Say “on request”	Deposit or public dataset	Shared when asked
74%	29%	9.8%	6.8%
2025, from 14% in 2015	2025, from 0.6%	2025, from 0.7%	of authors, elsewhere



16,362 open-access stroke full texts, up to 1,500 a year · Gabelica 2022

12 Finding 5

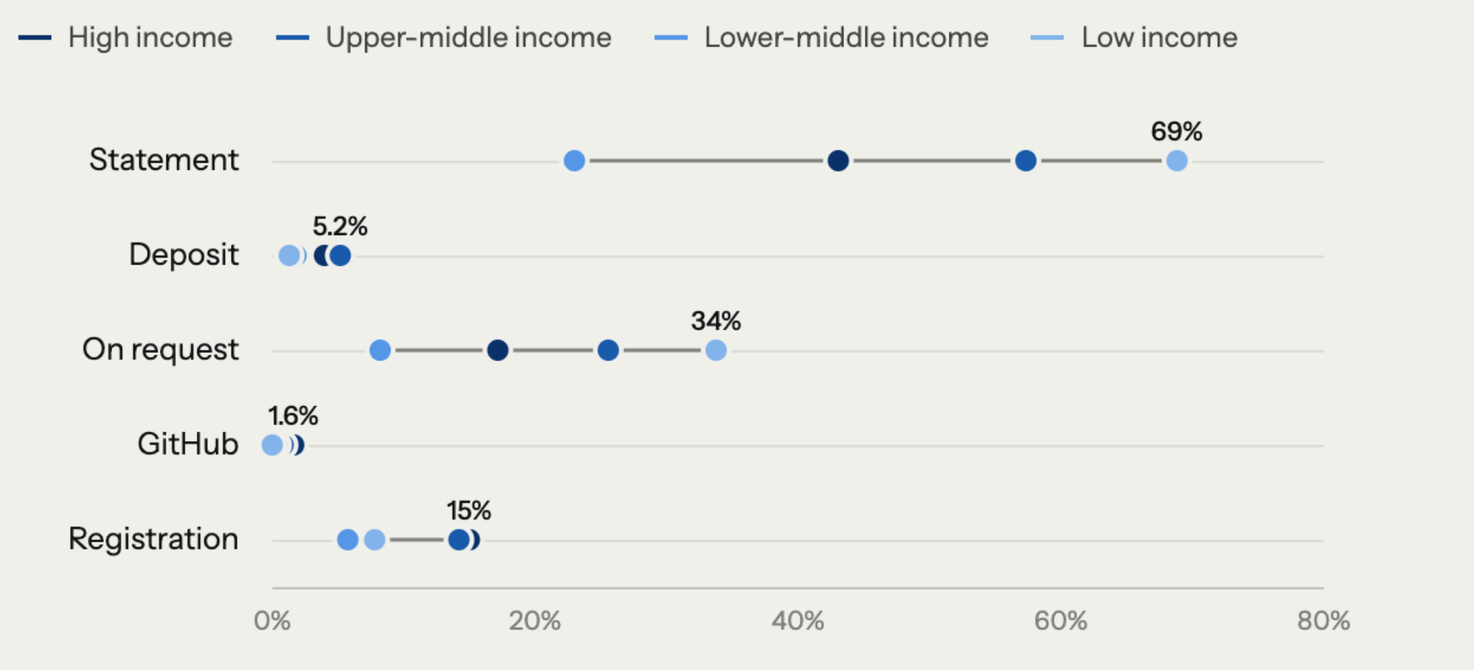
Income groups

42% vs 19%

free open-access routes, lower-middle vs high income

1.8% vs 4.0%

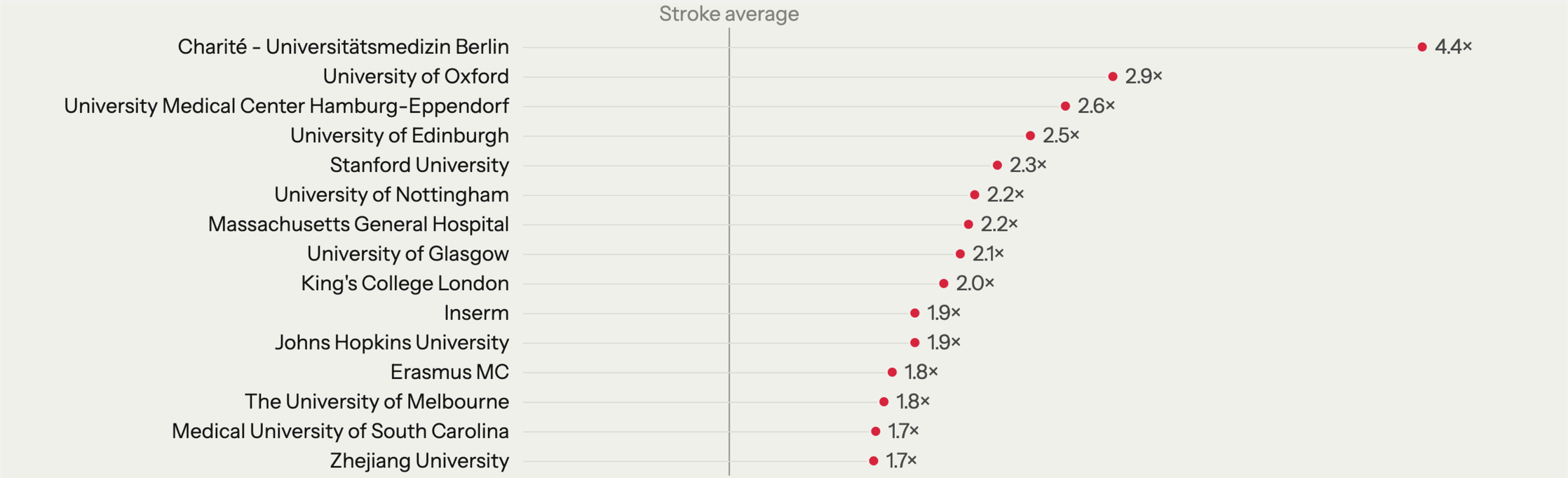
deposit or public data in the full text



World Bank income group of the first author's country, 2015–2025 · deposit from audited full texts

13 Finding 6

Commitment index



1 = stroke average · deposit, GitHub and registration in the full text · top 15 of the 60 institutions with most audited articles

14 Next

Incentives

3.1% → 3.1%	data obtainable, before and after the ICMJE rule	Bergeat 2022
0 of 14	stroke and neurology journals offer Registered Reports	COS list 2026
0 of 92	promotion guidelines mention open science	Rice 2020
44% vs 96%	positive results, Registered Reports vs standard	Scheel 2021

Our incentives point the other way.

15 Next

Proposals

1

Task force

WSO, on open science

- A position statement
- Retire “available on request”
- Preregistration beyond trials

2

Toolkits

WSO-backed, open, maintained

- Imaging, tested where most strokes happen
- Trial templates, data elements, consent
- This benchmark, for anyone

3

Prize

For the teams that share

- Commitment index, every six months
- Promotion criteria: 0 of 92 today

Targets 25% deposited data by 2030 · “on request” under 10% · non-trial registration 10% by 2030

WORLD STROKE CONGRESS 2026 · SEOUL

Collaborate

Stroke Open-Science Benchmark

friseeb.github.io/stroke_open_science_outputs

Explore it, challenge it, run it on your field

Code and data

github.com/Friseeb/stroke_open_science

Open source; the next run is January 2027

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Write to join the task force or to audit your own field

Disclosures: none.